MASTER SPECIFICATION | PRODUCTION GRADE | VIBRANT AMBER ACCENT | RELEASE V1.0.0

RandomTalks Pro

Executive Technical Architecture, Design System Tokens, WebRTC Mesh Signaling, ACID Matchmaking Protocol & Generative Intelligence Engine Specification.

ARCHITECTURE	DATA PLANE	AI ENGINE	TARGET COMPLIANCE
Clean HUD + WebRTC	Firestore Transactions	Gemini 1.5 Flash	Strict 18+ Verified

01. Executive Summary & High-Trust Paradigm

RandomTalks Pro redefines spontaneous video communication by eradicating the anonymity, toxicity, and latency characteristic of legacy random-chat platforms. Built on a clean *Pure White / Solid Amber HUD*, the platform enforces high-trust interpersonal networking through mandatory legal age attestation, structured purpose modes, mutual demographic compatibility, and automated AI icebreakers.

< 180ms

Signaling Latency

P2P ICE candidate negotiation using optimized STUN mesh pools.

100% ACID

Atomic Matching

Firestore transaction locks prevent race conditions & double joins.

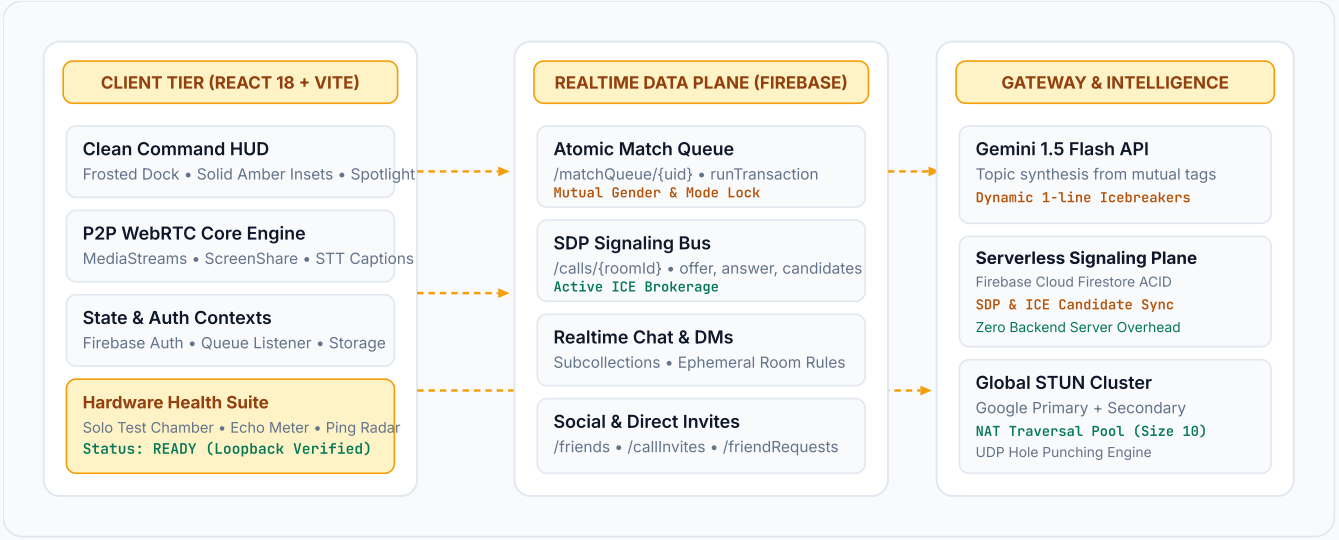
0% Vanity

Verified Context

Profile occupation badges, explicit interest tags, & mutual blocklists.

02. End-to-End System Architecture

SYSTEM TIER & PROTOCOL TOPOLOGY DIAGRAM



ASCII TOPOLOGY & TRANSPORT BLUEPRINT

SYSTEM MESH v1.0

```
[CLIENT A: BROWSER] ──(WebRTC P2P DTLS-SRTP / AV1)── [CLIENT B: BROWSER]
       |                                                    |
 SDP Signaling                                        SDP Signaling
       ▼                                                    ▼
┌──────────────────────────────────────────────────────────┐
| FIREBASE REALTIME PLANE (Firestore ACID Transactions)    |
|  • /matchQueue/{uid} ── Mutually filtered atomic locks   |
|  • /calls/{roomId}   ── Ephemeral SDP offers, answers & ICE |
└──────────────────────────────────────────────────────────┘
       |                                                    |
  Direct P2P DTLS-SRTP                              Dynamic Gemini Prompt
       ▼                                                    ▼
┌──────────────────────────────┐          ┌──────────────────────────┐
| DIRECT P2P DTLS-SRTP ENCRYPTION |       | GEMINI 1.5 FLASH ENGINE  |
|  • Pure serverless architecture |       |  • Sub-second icebreakers |
|  • Zero-vendor media transit    |       |  • 5 witty queue rules    |
└──────────────────────────────┘          └──────────────────────────┘
```

03. Design System & Dashboard Ergonomic Specification

The user interface implements the production Dashboard theme: **Clean White Surfaces with Solid Amber (#F59E0B) Focal Points**. It prioritizes clarity, readable contrast, and a floating frosted dock that keeps high-definition WebRTC video front and center.

TOKEN NAME	HEX / CSS VALUE	ROLE & ARCHITECTURAL APPLICATION
`bg-canvas`	`#FFFFFF / #F8FAFC`	Pure clean canvas backdrop. Replaces dark backgrounds across all views.
`bg-surface`	`#FFFFFF`	Crisp white surfaces, modals, drawer sidebars, and command HUD containers.
`border-card`	`#E2E8F0`	Subtle 1px perimeter boundary for cards and divider lines.
`accent-amber`	`#F59E0B / #D97706`	Solid Vibrant Amber. The sole primary brand accent for CTAs, active tags, and badges.
`text-main`	`#0F172A`	Deep Ink Charcoal. Used for primary headings, body copy, and high-contrast labels.
`text-muted`	`#64748B`	Neutral Slate. Used for secondary metadata, sub-labels, and explanatory captions.
`accent-emerald`	`#059669`	Emerald. Verified identity badges, latency indicator (<40ms ping), and healthy loopback.
`accent-danger`	`#DC2626`	Rose Red. Destructive actions: Disconnect, Block, Report, and Mic Mute active.

› Floating Dock Ergonomics

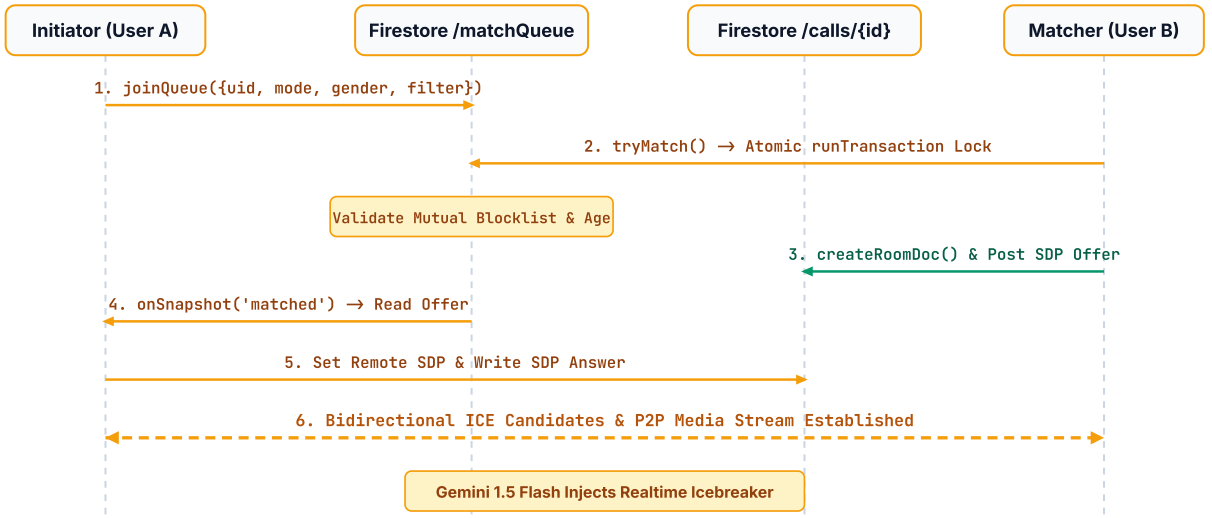
Controls are decoupled from the video stream into a floating frosted dock (`backdrop-blur-xl`) anchored 24px above the bottom viewport. Prevents accidental UI obscuration of partner facial cues.

› GPU Dedicated Decoupling

WebGL background presence meshes unmount instantly upon call handshake. 100% of browser hardware acceleration is allocated directly to NVDEC/AV1 video decoding.

04. ACID Matchmaking & WebRTC Signaling Protocol

■ SIGNALING SEQUENCE & STATE HANDSHAKE



› Matchmaking Algorithmic Guarantees

› **Dual-Condition Mutual Filter:** Candidate A is selected for Candidate B if and only if `(A.gender == B.genderFilter || B.genderFilter == 'Any')` *AND* `(B.gender == A.genderFilter || A.genderFilter == 'Any')`.

› **Anti-Griefing Blocklist Enforcement:** The Firestore transaction queries the caller's private `/blocks/{uid}/blockedUsers` index before committing, completely discarding matched candidates who have blocked or been blocked.

› **Stale Entry Eviction (Heartbeat Protection):** Queue candidates whose `joinedAt` timestamp exceeds 120 seconds are automatically evicted during transaction scans to purge dead browser tabs.

05. Production Firestore Data Schema Specification

The database plane strictly enforces structured schemas, ACID transaction guarantees, and subcollection encapsulation.

COLLECTION / PATH	PRIMARY FIELDS	DATA TYPES & INVARIANTS	ACCESS CONTROL RULE
`users/{uid}`	username, age, gender, genderFilter, occupation, interests, isVerified, createdAt	`string, number (⩾18), array, boolean, timestamp`	Owner Read/Write. Public read for profile card inspection.
`matchQueue/{uid}`	uid, username, gender, genderFilter, mode, status ('waiting'|'matched'), roomId	`string, enum, timestamp`	Authenticated Read/Write. Transaction-locked write.
`calls/{roomId}`	roomId, participants [uid1, uid2], callerOffer, calleeAnswer, status, createdAt	`string, string[], map (SDP), timestamp`	Participants only: `request.auth.uid in resource.data.participants`.
`calls/{roomId}/messages/{id}`	senderId, senderName, text, timestamp	`string, string, string (max 1000 chars), timestamp`	Enforced room participants only. Ephemeral life cycle.
`friends/{friendshipId}`	users [uid1, uid2], userMap {uid: {username, occupation}}, createdAt	`string[2] (lexicographically sorted), map, timestamp`	Member access only. Indexed for direct friend list querying.
`callInvites/{receiverUid}`	callerUid, callerName, receiverUid, roomId, status ('pending'|'declined'), createdAt	`string, string, enum, timestamp`	Direct call signaling. Trigger for Incoming Call Audio HUD modal.
`reports/{reportId}`	reporterUid, reportedUid, reason, notes, roomId, createdAt	`string, string, enum (harassment, under18, etc.), timestamp`	Create-only by authenticated users. Read restricted to Admin panel.

FIRESTORE SCHEMA DOCUMENT DEFINITION

JSON-SCHEMA v4

```
// Document Path: /calls/{roomId}
{
  "roomId": "usr_98a7f_usr_12bc4",
  "participants": ["uid_alice_98a", "uid_bob_12b"],
  "status": "active", // enum: 'waiting' | 'active' | 'ended'
  "callerOffer": { "type": "offer", "sdp": "v=0\r\no=- 42... IN IP4 0.0.0.0" },
```

```
  "calleeAnswer": { "type": "answer", "sdp": "v=0\r\no=- 87... IN IP4 0.0.0.0" },
  "mode": "tech", // 'tech' | 'casual' | 'friends' | 'deeptalk'
  "aiIcebreaker": "What's an exciting project or tech stack you're currently exploring?",
  "createdAt": ServerTimestamp()
}
```

06. Component Blueprints & Behavioral Matrix

MODULE NAME	SOURCE LOCATION	KEY RESPONSIBILITIES & CAPABILITIES
`CallRoom.jsx`	`src/pages/CallRoom.jsx`	Main WebRTC orchestration hub. Handles SDP signaling, dual camera feeds, floating frosted dock, 5-min timer chime, screen sharing, STT live captioning, and error boundaries.
`SoloTestModal.jsx`	`src/components/SoloTestModal.jsx`	Pre-flight hardware diagnostics. Live audio visualizer meter, camera aspect test, loopback audio test, and STUN connectivity verification prior to entering queue.
`ChatPanel.jsx`	`src/components/ChatPanel.jsx`	Real-time in-call text conduit. Supports markdown formatting, unread badge counters, instant scrolling, and participant auto-scoping.
`CallHistoryModal.jsx`	`src/components/CallHistoryModal.jsx`	Persistent connection history ledger. Tracks duration, partner metadata, private collaborative quick notes, and timestamp metrics.
`gemini.js`	`src/lib/gemini.js`	Gemini 1.5 Flash API connector. Generates context-aware conversational icebreakers and queue waiting etiquette tips with instant offline fallback pools.
`moderation.js`	`src/lib/moderation.js`	Instant user blocklist management, report dispatching, and safety verification ensuring high-trust platform standards.

07. Security, Trust & Threat Mitigation Matrix

Threat Vectors & Countermeasures

- **Client-Side Race Conditions:** Countered by Firestore `runTransaction` which rolls back if a candidate's status flips during matching.
- **Underage Participation:** Explicit legal confirmation checkbox required at profile creation. Under-18 accounts are rejected at client validation and blocked in firestore rules.
- **Zero-Vendor Privacy & Secret Isolation:** Direct browser-to-browser media streams are encrypted end-to-end via DTLS-SRTP. No third-party media servers (ZegoCloud/Agora) exist in the data path. Client authentication is strictly managed via Firebase Auth tokens.
- **P2P Media Privacy:** Direct WebRTC streams route peer-to-peer over DTLS-SRTP encryption; no intermediate video recording or cloud server interception occurs.

Production Deployment Checklist

- **Frontend:** React 18, Vite 5, Tailwind CSS, Lucide React icons.
- **Backend Infrastructure:** 100% Serverless Firebase Authentication and Cloud Firestore (Zero server maintenance costs).
- **Infrastructure:** Firebase Authentication (Email/Password), Cloud Firestore.
- **Turn / ICE:** Google Public STUN Cluster (5 geographically distributed endpoints).
- **AI Gateway:** Generative Language API endpoint (Gemini 1.5 Flash).